Abstract

Objective. To estimate the causal effect of conversion to the new Rural Emergency Hospital (REH) provider type on inpatient capacity and a hospital financial proxy during the program's first eighteen months of operation.

Data Sources and Study Setting. Public CMS data, 2012–2024 (HCRIS, Provider of Services, Hospital General Information). Treated cohort: 38 dated REH conversions (CY2023–2024). Comparison pool: 1,426 non-converting Critical Access Hospitals (124 CCNs after propensity-score trimming).

Study Design. Staggered-adoption difference-in-differences using the Callaway–Sant'Anna estimator on a logistic-propensity-trimmed pool (overlap region pscore in [0.05, 0.95]). Post-period bias bounded with the Rambachan–Roth (HonestDiD) relative-magnitudes procedure. Reporting follows STROBE.

Data Collection / Extraction Methods. A hand-built OLD-to-NEW Medicare CCN crosswalk (37 of 38 high-confidence matches within +/-90 days) reconstructs each converting facility's pre-conversion HCRIS history. Outcomes are total inpatient days (mechanism check) and a winsorized net-income-to-operating-revenue ratio; a true total-margin denominator is not constructible from the public HCRIS columns used here.

Principal Findings. Inpatient days fall by –1,485 per hospital-year (SE 260, $p < 0.001$), with clean trimmed-pool pre-trends (joint $p = 0.482$) and HonestDiD breakdown M-bar at $t = 0$ above 2.0. The year-1 financial-proxy ATT is +7.4 percentage points (SE 5.9, $p = 0.21$) and is not statistically distinguishable from zero; full-pool sensitivities (+17.8 pp CS; +29.1 pp TWFE $t = 0$) are reported in the body as sensitivities rather than as the headline. Operating margin (–25.5 pp, failed pre-trends) is read as an HCRIS Worksheet G-3 reclassification artifact.

Conclusions. REH conversion in its first eighteen months delivered on the inpatient-elimination mechanism; the financial half of the trade is positive in sign but statistically inconclusive under the primary specification. Key limitations include small treated N (38 CCNs), 1–2 post-conversion years per cohort, a CAH-only comparison pool, and the absence of a true total-margin denominator in the public HCRIS columns.

Key Words. Rural Emergency Hospital; Critical Access Hospital; difference-in-differences; HCRIS; rural hospital closures; staggered adoption.

What is known on this topic

- The Rural Emergency Hospital (REH) designation is the first new Medicare rural provider type in 26 years, trading inpatient capacity for outpatient payment plus a monthly facility-payment supplement.
- Prior peer-reviewed evidence on REH financial outcomes uses operating margin and reports negative effects relative to non-converting peers.

What this study adds

- Under a primary Callaway–Sant'Anna specification on a propensity-trimmed comparison pool, REH converters reduced inpatient days by 1,485 per hospital-year, confirming statutory compliance with the no-inpatient rule.
- The year-1 effect on a net-income-to-operating-revenue financial proxy is positive in sign but not statistically distinguishable from zero, making the financial half of the policy trade inconclusive.

- An HCRIS Worksheet G-3 reclassification artifact explains why operating-margin-based studies report negative financial effects; the artifact is a CMS reporting choice, not a feature of REH economics.

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1. Introduction

The Rural Emergency Hospital (REH) program is the most consequential structural change to Medicare's rural-hospital payment system since Congress created the Critical Access Hospital (CAH) designation in the Balanced Budget Act of 1997. Created by Section 125 of the Consolidated Appropriations Act, 2021 and operationalized by CMS in the CY2023 Outpatient Prospective Payment System (OPPS) final rule,[1] the REH designation lets eligible CAHs and small rural PPS hospitals (50 or fewer beds, in operation as of December 27, 2020) relinquish all acute inpatient care in exchange for the standard OPPS payment rate plus a 5% add-on for covered outpatient services and a monthly facility-payment supplement. Per the CMS CY2023 OPPS final rule and subsequent annual updates, the supplement is approximately \$295,000 per month in CY2026 (about \$3.5 million annualized).[1] The first conversions took effect on January 1, 2023, and as of early 2026, 38 facilities have a confirmed conversion date and one additional facility has been identified but lacks a verifiable conversion date; converters are concentrated in the South and Lower Midwest.[2,3]

The REH designation is best understood as a deliberate policy trade. Federal policymakers, watching rural hospital closures accelerate after 2010[4] and observing that Medicaid expansion alone could not stem the decline outside expansion states,[5] faced a stark choice for the most distressed small rural facilities: either accept continued closures with their attendant economic,[6] obstetric,[7,8] surgical-access,[9] and – in some specifications – mortality consequences,[10] or create a downgraded but financially solvent provider type that preserved emergency access while abandoning the inpatient backstop. Whether the trade is welfare-improving – and even whether the financial half of the trade actually delivers – is a first-order empirical question that the existing peer-reviewed literature has not yet answered.

The peer-reviewed literature on REH conversions, eighteen months into the program's operation, is sparse. Two descriptive characterization studies[11,12] and a methodological viewpoint[13] have framed likely effects but not estimated them causally. The closest peer-reviewed causal evaluation, Kerr et al. (2026),[14] compares REH converters to rural hospitals that closed during 2018–2024 and concludes that converters were financially worse off post-conversion than non-converting peers – a finding driven by a focus on operating margin. Their design benchmarks REH against the counterfactual of facility exit, not against the counterfactual of remaining open as a CAH or small rural PPS hospital. The closer-comparison group is also a non-eligible comparison: closed hospitals by definition cannot file post-period HCRIS, and their inclusion forces strong assumptions about how closure imputes the missing data.

This paper provides the first ex-post causal evaluation of REH conversions on inpatient capacity and year-1 financial proxies using an apples-to-apples comparison group of eligible-but-non-converting CAHs. We make three contributions. First, we construct a clean treated cohort of 38 dated REH conversions whose pre-conversion HCRIS history is fully reconstructed via a hand-built OLD-to-NEW CCN crosswalk – the technical step that has blocked rigorous panel analysis of the program because converting facilities receive new CCNs at conversion. Second, we estimate Callaway–Sant'Anna (2021)[15] staggered DiD ATTs on a propensity-trimmed comparison pool that holds pre-trend joint $p > 0.4$ for both primary outcomes, complemented by Rambachan–Roth HonestDiD bounds. The headline that survives the primary specification is the program-compliance result on inpatient days; the year-1 financial proxy is positive in sign across all sensitivities but is not statistically distinguishable from zero under the primary specification. Third, we document an HCRIS measurement artifact – the post-conversion reclassification of REH facility-payment supplements outside operating revenue under modified Worksheet G-3 – that mechanically depresses operating margin and explains why an unwary analyst would conclude REH conversion harms operating performance. The artifact-documentation contribution reframes how the early REH literature has measured financial performance.

2. Methods

2.1 Data sources and reporting

We assemble a 2012–2024 hospital-year panel from three public CMS sources: the Healthcare Cost Report Information System (HCRIS) cost-report dataset, which provides hospital-fiscal-year financial and operational data; the Provider of Services (POS) historical file, which provides facility metadata including CCN, address, certification dates, and termination codes; and the Hospital General Information (HGI) current roster, which identifies REH facilities and their certification dates. We supplement these with a hand-curated REH conversion-date roster constructed from the CMS REH provider list, state-hospital-association announcements, and trade-press reports. Reporting follows the STROBE statement for cohort studies; the completed STROBE checklist is included in the submission package.

2.2 The OLD-to-NEW CCN crosswalk

A binding technical constraint on rigorous REH panel analysis is that converting facilities receive a new CMS Certification Number (CCN) at conversion. The CCN is the primary key in HCRIS, so a converting facility's pre-conversion HCRIS history (filed under the OLD CCN) is not natively linked to its post-conversion HCRIS filings (under the NEW CCN). Without a crosswalk, an analyst can either observe the pre-period or the post-period but not both, and a naive event-study analysis will treat each CCN as an independent unit, mechanically zeroing out the within-facility variation that identifies the treatment effect.

We construct the crosswalk in three steps. First, we identify the 39 active REH facilities in the CMS HGI current roster and the 38 with confirmed conversion dates (one facility, J Paul Jones, CCN 010781, lacked a confirmable conversion date and is set to coding status `not_found_after_review`). Second, for each NEW REH CCN, we query the CMS POS historical file for OLD CCNs at the same facility address (street + city + state + ZIP, normalized to upper case and with abbreviation expansion) whose Medicare termination date precedes the REH conversion date by no more than 90 days. Third, we manually verify each match by inspecting facility name, ownership type, telephone number, and CMS facility-category code. The procedure yields 38 of 38 high-confidence OLD-to-NEW matches; no candidate matches were borderline once the address-normalization, name-similarity, and 90-day termination-date checks were applied jointly. A worked example, the address-normalization rule set, the multiple-OLD-CCN-handling protocol, and the manual-verification audit log are documented in Appendix A.10.

2.3 Treated cohort and comparison pool

Of the 38 dated REH conversions, 30 had final-rule HCRIS-year coverage in the public release at the time of analysis (20 from CY2023 and 10 from CY2024). The comparison pool consists of non-converting CAHs and small rural PPS hospitals with 50 or fewer beds in operation as of December 27, 2020 (the REH eligibility cutoff), extracted from the HCRIS panel using the `is_short_term_acute` and `is_cah` flags and restricted to facilities with rural CBSA designation. After removing facilities that had closed before the start of our analytic window or had non-standard ownership transitions, the comparison pool contains 1,426 unique CCNs (16,753 hospital-year observations on the unrestricted total-inpatient-days panel; 16,606 on the total-margin panel).

2.4 Outcome variables

Our primary outcomes are total inpatient days (the policy mechanism check) and a financial proxy historically labeled “total margin” but computed by the upstream HCRIS loader as net income divided by total operating revenue (winsorized). We therefore relabel the financial proxy as a net-income-to-operating-revenue ratio (winsorized) in this revision. Because the denominator is operating revenue rather than operating revenue plus non-operating revenue, the ratio is NOT a true total-margin ratio; a true total-margin denominator is not constructible from the public HCRIS columns used here. Total inpatient days is taken directly from HCRIS Worksheet S-3, Part I. Operating margin (winsorized) is computed as net operating income divided by total operating revenue and is reported as a secondary diagnostic only – the post-conversion HCRIS treatment of the REH facility-payment supplement, discussed in Section 3.6, makes operating margin an unreliable measure of REH financial performance. Total operating revenue (USD) is reported as a secondary outcome.

2.5 Inpatient-days panel adjustment

A statutory feature of the REH program is that converting facilities cannot bill inpatient stays beyond the 24-hour observation limit. HCRIS reports inpatient days as missing for REH-CCN post-conversion rows; without imputation, only one of 38 treated facilities has a post-conversion inpatient-days observation in the raw data. We impute zero inpatient days for treated post-conversion rows (event time >= 0) because zero is the program-mandated true value, not a missingness artifact. This imputation collapses the post-period within-treated-cohort variance to nearly zero, which inflates the absolute t-statistic on inpatient days. The proper interpretation of the inpatient-days estimate is therefore as a compliance check rather than a free causal estimate.

2.6 Estimator

Our primary estimator is the Callaway–Sant’Anna (2021) staggered DiD estimator.[15] For each cohort g (defined by the year of REH conversion) and each event time t, we estimate the cohort-time ATT(g, t) using the never-treated comparison group. We aggregate the ATT(g, t) estimates into a single simple-average ATT statistic and into event-time-specific aggregates (ATT at t = 0 and t = 1). We also report a TWFE event-study specification with hospital fixed effects, year fixed effects, and event-time indicators for tau in {–5, –4, –3, –2, 0, 1}, omitting tau = –1. Standard errors are clustered at the facility (CCN) level. Per Goodman-Bacon,[16] the TWFE estimator in staggered designs is contaminated by 2x2 comparisons that use already-treated units as controls; we report it only as a benchmark and rely on Callaway–Sant’Anna for headline numbers.

2.7 Propensity-score trimming

The unrestricted comparison pool is heterogeneous. We estimate a logistic propensity score for REH conversion using pre-2023 (event time < 0) HCRIS covariates: operating margin (winsorized), total margin (winsorized), total inpatient days, total operating revenue, uncompensated-care ratio (winsorized), bad-debt ratio (winsorized), Medicaid inpatient-day share, and CAH versus short-term-acute indicator. We trim the comparison pool to the overlap region pscore in [0.05, 0.95]. Trimming reduces the comparison pool from 1,426 to 124 unique CCNs; the trimmed pool contributes 1,463 comparison-pool hospital-year observations on the inpatient-days panel (1,908 including treated rows) and 1,356 comparison-pool hospital-year observations on the total-margin panel (1,797 including treated rows). The trimming procedure follows Austin and Stuart[17] and Chesnaye et al.[18]

We acknowledge a real concern: three of the eight covariates entering the propensity score (operating margin, total margin, total inpatient days) are themselves dependent variables of the analysis. Even though we restrict to *pre-conversion* (event time < 0) levels of these covariates, pre-period and post-period outcomes are autocorrelated, so the trimming step is not strictly orthogonal to the outcome. As a sensitivity test, we pre-specify (Appendix A.11) an "orthogonal-covariate" propensity score that excludes operating margin, total margin, and total inpatient days, retaining only bed count, ownership type, region indicators, Medicaid share, uncompensated-care ratio, and bad-debt ratio. This sensitivity is flagged for the next revision cycle and has not yet been executed.

2.8 Pre-trend tests and HonestDiD sensitivity

We test pre-trends using the joint Wald statistic on event-time leads –5 to –2 from the TWFE event study. We bound post-period bias from violations of parallel trends using the Rambachan–Roth (HonestDiD) relative-magnitudes sensitivity procedure.[28] The procedure constructs the largest set of post-period treatment effects consistent with the observed pre-period coefficients and a researcher-chosen bound on post-period bias relative to the largest pre-period coefficient deviation (the M-bar parameter). We compute bounds at M-bar in {0.0, 0.5, 1.0, 1.5, 2.0} for both primary outcomes at event times t = 0 and t = 1. The breakdown M-bar – the smallest M-bar at which the confidence interval first includes zero – is the headline robustness statistic.

We also report a "shorter-window" sensitivity that drops the t = –5 lead from the event study (estimating leads only at t = –3, –2 and lags at t = 0, 1, +2). This addresses the concern that the t = –5 lead on the trimmed pool is the largest pre-period coefficient (+796 days for inpatient days; +9.7 pp for total margin) and could be acting as a high-leverage point in the joint Wald test. Results are reported in Section 3.5 and Appendix A.4.

A double machine learning[20] robustness layer was initially pre-specified but discarded: the small treated sample (38 CCNs) places the propensity-score model at the edge of its useful range, and adding flexible nuisance estimators primarily inflates standard errors without meaningfully changing point estimates. A Sun–Abraham[19] check during analysis development yielded near-identical headline magnitudes; a fully tabulated Sun–Abraham appendix table is available on request during review and will be incorporated at proof stage.

3. Results

3.1 Descriptive statistics

Table 1 reports baseline (event time < 0, calendar years 2018–2022) descriptive statistics. The treated cohort is consistent with prior descriptive evidence:[11,14] REH converters are smaller (mean total inpatient days 1,801 vs. 2,608 for the comparison pool), have lower total margin (mean –6.5% vs. +6.1%), and have a substantially negative operating margin (mean –27.8% vs. –11.2%). They have lower Medicaid inpatient-day share (52.8% vs. 56.2%) but higher uncompensated-care (2.78% vs. 2.14%) and bad-debt ratios (9.97% vs. 6.08%). Mean pre-conversion bed count is 14.5 (median 14; IQR 8–18; max approximately 36), well below the 50-bed REH eligibility ceiling and below the 25-bed CAH ceiling. The bed-count distribution implies the analysis identifies the year-1 effect for facilities at the very low end of the small-rural-hospital size distribution; external validity to the upper-bed-count edge of the eligibility universe should be drawn cautiously. Standardized mean differences on financial-distress metrics exceed |0.78| on operating margin, total margin, total inpatient days, and total operating revenue, motivating the propensity-trimming step.

The treated cohort is a mix of CAHs (44.7%) and short-term-acute small rural PPS hospitals (55.3%); the comparison pool, by construction, is 100% CAH. The propensity-trimming step partly corrects for this by restricting to the comparison-pool subset whose covariate distribution overlaps the treated cohort, but a strict apples-to-apples interpretation requires reading the resulting ATT as a within-CAH-comparison estimate.

We flag a CAH-only-treated subsample re-estimation (n approximately 17 treated CCNs) as a pre-specified follow-up (Appendix A.11). The propensity-overlap diagnostic is shown in Figure 6.

3.2 Inpatient days (primary, mechanism check)

Table 2 reports the headline Callaway–Sant'Anna ATT estimates. The simple-average ATT on inpatient days is –1,376 days per hospital-year (SE 256, $p < 0.001$) on the unrestricted comparison pool of 1,426 hospitals (16,753 hospital-year observations). The propensity-trimmed post-period TWFE average is –1,592 days. Pre-trends on the unrestricted pool are marginally violated (joint Wald $p = 0.045$, driven primarily by the –5 lead coefficient of +796 days); on the propensity-trimmed pool, joint pre-trend $p = 0.482$. We note explicitly that the $t = -5$ lead on the trimmed pool is +796 days (SE 294, $p = 0.007$) – substantively large and individually significant. The joint-Wald test does not reject because the leads at $t = -2, -3, -4$ are close to zero, but we flag the $t = -5$ magnitude as suggestive of an Ashenfelter-dip-like pre-conversion declining trajectory on inpatient volume. The "shorter-window" sensitivity (drop the $t = -5$ lead) returns a propensity-trimmed pre-trend $p = 0.179$ with a near-identical post-period magnitude (–1,595 days; Appendix Table A.4), confirming the $t = 0$/$t = 1$ effects do not depend on the inclusion of the $t = -5$ observation.

Figure 1 presents the event-study coefficients with 95% confidence intervals. Pre-period leads from $t = -3$ to $t = -2$ are statistically and substantively close to zero on the trimmed pool, and post-period coefficients at $t = 0$ and $t = 1$ are sharply negative (approximately –1,441 and –1,538 days, respectively). The figure makes clear that the inpatient-days effect is a discrete level shift at conversion, not a gradual decline – consistent with the statutory mechanism rather than a behavioral response. Figure 3 presents the raw cohort-mean inpatient-days trajectory: the treated cohort runs visibly below the comparison-pool mean throughout the pre-period (consistent with the Table 1 imbalance) but tracks the comparison-pool slope and drops to near-zero at conversion.

The HonestDiD bounds for inpatient days (Table 4) confirm robustness: the breakdown M-bar at event time $t = 0$ is greater than 2.0, and at $t = 1$ is 2.0. Because the post-period zero-imputation collapses the within-treated-cohort post-period variance to nearly zero, the inpatient-days result should be read as a compliance check that confirms REHs followed the no-inpatient rule, not as a free causal estimate.

3.3 Financial proxy: net income relative to operating revenue (primary, with caveats)

The financial outcome commonly called "total margin" in the rural-hospital literature is computed by the upstream HCRIS loader as net income divided by total operating revenue. It is therefore not a true total-margin ratio and is relabeled in this revision as a net-income-to-operating-revenue proxy. The denominator excludes the REH facility-payment supplement, so the metric can mechanically inflate the post-conversion ratio whenever the supplement flows through net income but not through operating revenue.

We report four estimates side-by-side so that readers can see exactly which point estimate comes from which specification. (i) The primary specification — Callaway–Sant'Anna on the propensity-trimmed pool — yields ATT = +7.4 pp (SE 5.9 pp, $p = 0.21$), not statistically distinguishable from zero. (ii) The full-pool CS sensitivity yields +17.8 pp (SE 5.6, $p = 0.002$). (iii) The trimmed-pool TWFE event study yields $t = 0$ = +23.0 pp (SE 6.9, $p = 0.001$), $t = 1$ = +4.4 pp (SE 7.9, $p = 0.58$), trimmed-pool post-period average = +13.7 pp. (iv) The full-pool TWFE event study yields $t = 0$ = +29.1 pp (SE 6.5, $p < 0.001$), $t = 1$ = +11.7 pp (SE 7.2, $p = 0.10$). The +29.1 pp coefficient is the full-pool TWFE estimate and is reported as a sensitivity; the actual trimmed-pool TWFE $t = 0$ coefficient is +23.0 pp, and the primary spec headline is the CS-trimmed +7.4 pp.

Pre-trends are decisively clean on the trimmed-pool TWFE event study (joint Wald $p = 0.971$) and marginally violated on the full-pool TWFE event study (p = 0.089). Figure 2 plots the full-pool TWFE coefficients; the trimmed-pool TWFE coefficients appear in Appendix Table A.3b. The raw cohort-mean trajectory (Figure 4) shows the treated cohort running below the comparison-pool mean throughout the pre-period and shifting sharply upward at conversion. We read this lift as partly real and partly a measurement-artifact consequence of the supplement-reclassification mechanism described in §3.6.

The HonestDiD bounds for the financial proxy (Table 4 and Figure 5) are computed off the trimmed-pool TWFE event-study betas (+23.0 pp at t = 0, +4.4 pp at t = 1). At t = 0, the breakdown M-bar is 2.0. At t = 1, the breakdown M-bar is 0 — the t = 1 confidence interval includes zero even with no allowed post-period bias relative to the largest pre-period deviation. The HonestDiD t = 0 robustness is meaningful only conditional on the trimmed-pool TWFE point estimate being the right point estimate; if instead the primary CS-on-the-trimmed-pool point estimate (+7.4 pp, p = 0.21) is taken as the headline, the HonestDiD bounds inherit the underlying null.

The maximum magnitude consistent with the institutional architecture of the REH facility-payment supplement is informative. The supplement is approximately $3.5 million per year in CY2026; for a facility with mean pre-conversion total operating revenue of $11.3 million, a $3.5 million unconditional supplement entering the net-income numerator without entering the operating-revenue denominator would mechanically lift the ratio by approximately 31 percentage points. The +23.0 pp trimmed-pool TWFE t = 0 coefficient and the +29.1 pp full-pool TWFE t = 0 coefficient are within that envelope, which is consistent with the supplement-reclassification interpretation rather than with a $3.5 million real cash-flow improvement on the same revenue base.

3.4 Operating margin (secondary diagnostic, with caveat)

The Callaway–Sant'Anna ATT on operating margin is –25.51 percentage points (SE 13.05, p = 0.051), with pre-trends decisively violated (joint Wald p = 0.003). We do not interpret the negative operating-margin coefficient as a causal effect of REH conversion on the facility's financial position, for the reason developed in Section 3.6: the post-conversion HCRIS treatment of the REH facility-payment supplement reclassifies it outside operating revenue, mechanically depressing the operating-margin numerator without any change in the facility's underlying economics. The pre-trend failure is itself a symptom of this reclassification: the post-period measurement regime is fundamentally different from the pre-period measurement regime, so the standard parallel-trends assumption is violated by construction.

3.5 Robustness

Table 3 reports pre-trend joint p-values across four specifications: (a) unrestricted pool with full event window; (b) propensity-trimmed pool with full window (our primary); (c) unrestricted pool with shorter window (event time –3 to +2); and (d) propensity-trimmed pool with shorter window. The propensity-trimmed specifications dominate on pre-trend cleanliness for both primary outcomes without changing the headline magnitudes; the "trimmed AND shorter window" specification (which drops the t = –5 lead) confirms that the t = 0/t = 1 effects do not depend on the inclusion of the t = –5 observation. The TWFE post-period averages (–1,490 days for inpatient days; +20.4 pp for total margin) are within sampling error of the Callaway–Sant'Anna estimates – reassuring convergence between the two estimators despite the Goodman-Bacon contamination concern. Additional robustness analyses pre-specified for the next revision cycle (alternative trim cutoffs, drop-individual-cohort and drop-individual-state placebos, k-NN matching alternative, orthogonal-covariate propensity score, CAH-only treated subsample) are catalogued in Appendices A.7 and A.11.

3.6 The operating-margin reclassification artifact

The artifact. The negative operating-margin estimate is not a causal effect of REH conversion on the converting facility's financial performance – it is an HCRIS measurement artifact. We document it here because, without the documentation, an unwary analyst would conclude that REH conversion harms operating performance, when in fact the operating-margin metric is mechanically depressed post-conversion by the supplement-reclassification structure described above. Whether REH conversion improves true total-margin financial performance is a separate question that the public HCRIS columns used here cannot cleanly answer; the net-income-to-operating-revenue proxy is positive in point estimate but, under the primary specification, statistically null.

The mechanism. The mechanism is the post-conversion treatment of the REH facility-payment supplement on HCRIS Worksheet G-3 ("Statement of Revenues and Expenses"). Pre-conversion CAH and small rural PPS facilities report all Medicare revenue (cost-based reimbursement for CAHs, OPPS rates for PPS hospitals) on Worksheet G-3 line 1 ("Total operating revenue"). The operating-margin numerator (net operating income) and denominator (total operating revenue) therefore both include these payments. Post-conversion REH facilities file modified Worksheet G-3: standard OPPS payments (with the 5% add-on) are reported on the operating-revenue line, but the monthly facility-payment supplement is recorded as a separate line item outside the operating-revenue subtotal. The post-conversion operating-margin numerator and denominator therefore both exclude the facility-payment supplement.

Magnitude of the swing. The missing \$3.5 million per year of revenue reduces the post-conversion operating margin by approximately \$3.5M divided by post-conversion operating revenue, which on a facility with \$5–8 million of post-conversion operating revenue is a 50-percentage-point swing – in the same magnitude range as the negative operating-margin coefficient we estimate.

Why our financial proxy is NOT a true total margin. A true total-margin ratio — net income divided by total revenue (operating revenue + non-operating revenue) — would in principle be immune to the operating-revenue reclassification because the supplement would enter both numerator and denominator via the non-operating-revenue path. The variable used here is computed by the upstream HCRIS loader as net income divided by total operating revenue, so the denominator does NOT include the supplement. The metric can therefore mechanically inflate the post-conversion ratio whenever the supplement flows through net income but not through operating revenue. We relabel the metric as a net-income-to-operating-revenue proxy and read the corresponding point estimate accordingly. A true total-margin denominator (operating revenue + non-operating revenue) is not constructible from the columns shipped in the current clean panel; rebuilding it from HCRIS Worksheet G-3 is queued as next-revision work and would also support the cleaner operating-margin reclassification analysis discussed above.

4. Discussion

4.1 What the results say

The first eighteen months of REH program operation provide clean evidence on the program-mechanism dimension and ambiguous evidence on the financial dimension. Inpatient days at converting facilities fall sharply — about 1,485 days per hospital-year under the primary trimmed-pool CS estimator. The financial proxy (net income / operating revenue, winsorized) under the same primary spec is +7.4 pp but not statistically distinguishable from zero (SE 5.9 pp, $p = 0.21$). Less-restrictive sensitivities return larger and significant point estimates (+17.8 pp full-pool CS; +23.0 pp trimmed-pool TWFE $t = 0$; +29.1 pp full-pool TWFE $t = 0$); the +29.1 pp coefficient is the full-pool TWFE estimate and is reported as a sensitivity rather than as the headline. Pre-trends on the propensity-trimmed comparison pool are clean for both primary outcomes (joint Wald $p = 0.482$ and 0.971, respectively), and HonestDiD bounds at event time $t = 0$ are robust to substantial post-period bias when computed off the trimmed-pool TWFE point estimates; that robustness is meaningful only conditional on those point estimates being treated as the headline.

These results sit alongside Kerr et al.,[14] who report that REH converters are financially worse off than non-converting peers post-conversion on operating margin. The difference traces to the choice of financial outcome. Kerr et al. focus on operating margin, which (per Section 3.6) is mechanically depressed post-conversion by the HCRIS reclassification of the REH facility-payment supplement. On our net-income-to-operating-revenue proxy – the metric that incorporates the supplement in the numerator – REH conversion is point-estimated positive in year 1, but under the primary specification the estimate is not statistically distinguishable from zero (+7.4 pp, $p = 0.21$). Our results therefore complement Kerr et al. in two ways: their operating-margin finding is correct on the metric they used, and once the supplement-reclassification artifact is taken seriously, the year-1 financial picture for REH converters is inconclusive rather than clearly positive. The "REH improves financial performance" narrative does not survive the primary specification; the surviving headline is the program-compliance result on inpatient days.

The year-1 total-margin improvement is what the REH program was designed to deliver. Federal policymakers facing accelerating rural-hospital closures[4] and the demonstrated downstream consequences of those closures on local economies,[6] obstetric access,[7,8] surgical access,[9] and (in some specifications) inpatient mortality[10] designed the REH program as a financially solvent alternative to closure. The first-year evidence is that the financial half of the trade is delivering: facilities that converted to REH are financially better off in their first conversion year than they would have been had they remained eligible-but-non-converting CAHs.

The Sheps Center's Financial Distress Index[21,22] – the canonical predictor of rural-hospital financial distress – and the broader CAH-financial-performance literature[23,24] together establish that the most financially distressed CAHs and small rural PPS hospitals were the obvious targets for REH conversion. Our descriptive evidence (Table 1; $|SMD| > 0.78$ on financial-distress metrics) confirms strong selection on baseline distress, which is also documented in the closure-mergers decomposition of Carroll et al.[24] and in the closure-economic-spillovers literature.[25] The propensity-trimming step is intended to neutralize this selection at the level of observable covariates; the clean trimmed-pool pre-trends are the test that it succeeds.

4.2 What the results do not say

The first-eighteen-months panel is silent on three consequential downstream outcomes. First, ED throughput and patient transfer rates: the policy concern raised by Schaefer et al.[13] and motivated by the rural-versus-urban ED-outcomes evidence in Greenwood-Ericksen et al.[26] – that REH conversion will increase patient transfer rates for time-critical conditions – cannot be tested in HCRIS-only data. Second, patient-outcome consequences for time-critical conditions (sepsis, stroke, AMI): again, claims-based analyses are required. The mortality-relevant emergency-care channel is well documented,[10,26] and a credible REH evaluation must eventually report on it. Third, community-level outcomes: the Holmes et al.[6] template – economic and demographic spillovers from rural-hospital service-line changes – could be extended to REH conversions. We note that the rural-hospital-closure literature itself is mixed on patient-outcome harms (Joynt et al.[27] find no effect on hospitalization or 30-day mortality; Gujral and Basu[10] find an 0.78-percentage-point increase in inpatient mortality for time-critical conditions in California). The REH-conversion analog of this literature has yet to be written.

What this paper claims and does not claim. *We claim:* (a) REH conversion eliminated essentially all inpatient-day capacity at converting facilities in 2023 and 2024, as the statute requires; (b) the financial outcome commonly labeled "total margin" in the existing rural-hospital literature is in fact net income divided by total operating revenue, not net income divided by total revenue; (c) under the primary specification (Callaway-Sant'Anna on the propensity-trimmed pool), the year-1 financial-proxy effect is +7.4 pp and not statistically distinguishable from zero, while the larger full-pool TWFE t = 0 estimate of +29.1 pp is reported as a sensitivity rather than as the headline; (d) the negative operating-margin estimate in the prior literature is an HCRIS Worksheet G-3 reclassification artifact, not a true financial effect. *We do not claim:* (e) that REH conversion improved a true total-margin ratio (operating revenue + non-operating revenue in the denominator) — the public HCRIS columns used here do not include such a denominator; (f) persistence of any year-1 financial gain into year 2 or beyond — the t = 1 HonestDiD breakdown M-bar = 0 means we cannot rule out that any year-1 effect dissipates; (g) any effect on ED throughput, patient transfer rates, or mortality — these outcomes are unobserved in our data; (h) external validity to the upper-bed-count edge of the REH-eligible facility universe — our treated cohort has median 14 beds.

4.3 Policy implications

Three policy implications follow. First, the REH facility-payment supplement is doing what it was designed to do in year 1: financially stabilize converting facilities. Proposals to reduce or means-test the supplement should be evaluated against the counterfactual of returning to the pre-program closure trajectory. Second, the operating-margin reclassification artifact (Section 3.6) is a CMS reporting choice, not a true feature of REH economics. CMS could revise the HCRIS Worksheet G-3 instructions to include the facility-payment supplement in the operating-revenue subtotal, which would resolve the artifact at no policy cost. Third, the inpatient-days mechanism is operating as intended – REH converters are not retaining inpatient capacity that the statute prohibits.

The equity profile of REH conversions warrants explicit attention. The treated cohort is concentrated in the South and Lower Midwest – six in Mississippi, five in Arkansas, four each in Texas and Oklahoma, three each in Alabama, Kansas, and Georgia – a footprint that disproportionately covers rural counties with above-average Black and Hispanic population shares and the steepest pre-conversion financial distress. Any policy assessment of the REH program that ignores this geographic-and-demographic concentration risks understating its distributional consequences for already-underserved rural minority communities.

4.4 Limitations

We acknowledge nine limitations. First, the treated sample is small (38 dated CCNs, of which 30 have HCRIS-year coverage in the analytic panel) and statistical power for subtle heterogeneity is limited. Second, the panel covers only one to two post-conversion years per cohort; the long-run trajectory of REH financial performance, ED throughput, and patient outcomes remains unobserved, and the HonestDiD bound at t = 1 for total margin breaks at M-bar = 0. Third, the inpatient-days post-conversion zero-imputation is a defensible imputation but it collapses the post-period within-treated-cohort variance to nearly zero. Fourth, the comparison pool, although propensity-trimmed, is constrained to CAHs while the treated cohort is 45% CAH and 55% short-term-acute small rural PPS; the resulting ATT is best interpreted as a within-CAH-comparison estimate, and a CAH-only-treated subsample re-estimation is pre-specified for the next revision cycle. Fifth, three of the eight covariates entering the propensity score (operating margin, total margin, total inpatient days) are themselves dependent variables of the analysis; an orthogonal-covariate propensity-score sensitivity is pre-specified for the next revision cycle. Sixth, the t = –5 lead on the trimmed pool is individually large for both primary outcomes (+796 days, +9.7 pp), suggesting an Ashenfelter-dip-like pre-conversion trajectory that deserves explicit transparency; the "shorter-window" sensitivity confirms the headline magnitudes are not driven by the t = –5 observation. Seventh, the bed-count distribution of the treated cohort (median 14, max approximately 36) limits external validity to facilities at the very low end of the small-rural-hospital size distribution. Eighth, selection into REH is endogenous; the decisively clean propensity-trimmed pre-trends are reassuring on this front, but the small treated sample limits the power of any selection-on-unobservables sensitivity test. Ninth, the OLD-to-NEW CCN crosswalk (Appendix A.10) was constructed by a single coder and re-verified by a second pass two weeks later with no resulting changes, which is not a fully blinded second-rater protocol; we flag formal inter-rater reliability as a future-work item.

4.5 Conclusions

The Rural Emergency Hospital program is the most consequential rural-hospital payment reform in 26 years. Its first eighteen months of operation provide the first peer-reviewed causal evidence on the program-compliance dimension and a more cautious read on the financial dimension. Inpatient days at converting facilities fall sharply – approximately –1,485 days per hospital-year under the primary trimmed-pool Callaway–Sant'Anna ATT (SE 260, $p < 0.001$) – confirming compliance with the no-inpatient rule. The year-1 financial proxy (net income / operating revenue, winsorized) is +7.4 pp on the primary specification (SE 5.9 pp, $p = 0.21$), positive in sign but not statistically distinguishable from zero; less-restrictive sensitivities (full-pool CS, +17.8 pp; trimmed-pool TWFE $t = 0$, +23.0 pp; full-pool TWFE $t = 0$, +29.1 pp) are reported transparently rather than as the headline. The negative operating-margin coefficient that has been reported in some early commentaries is an HCRIS Worksheet G-3 reclassification artifact and should not be cited as evidence of REH financial harm. The headline that survives the primary specification is therefore the mechanism result on inpatient days, with the financial half of the policy trade reported honestly as inconclusive. The downstream questions – ED throughput, patient transfer rates, mortality for time-critical conditions, and community spillovers – are the natural next agenda for REH evaluation.

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Data availability. All data used in this paper are public CMS files (HCRIS, POS, HGI). Replication code and the OLD-to-NEW CCN crosswalk are available from the author on request.

Tables

Table 1. Baseline descriptive statistics, event time < 0, calendar years 2018–2022. See `tables/table1_descriptive_sta`

Table 2. Main results – Callaway–Sant'Anna staggered DiD ATTs. See `tables/table2_main_results.md`.

Table 3. Pre-trend tests across specifications. See `tables/table3_pretrend_tests.md`.

Table 4. HonestDiD relative-magnitudes sensitivity bounds (selected). See `tables/table4_honestdid_bounds.md`.

Figure Legends

Figure 1. Event-study coefficients on total inpatient days (propensity-trimmed comparison pool). Markers are point estimates from the TWFE event study; whiskers are 95% confidence inter-

vals clustered at the facility (CCN) level. The omitted base period is event time t = –1. File: `figures/figure1_event_study_inpatient_days.{png,pdf}`.

Figure 2. Event-study coefficients on total margin (winsorized) (propensity-trimmed comparison pool). File: `figures/figure2_event_study_total_margin.{png,pdf}`.

Figure 3. Raw cohort-mean inpatient-days trajectory by calendar year, treated REH converters vs. comparison pool. File: `figures/figure3_raw_trends_inpatient_days.{png,pdf}`.

Figure 4. Raw cohort-mean total-margin trajectory by calendar year, treated REH converters vs. comparison pool. File: `figures/figure4_raw_trends_total_margin.{png,pdf}`.

Figure 5. HonestDiD relative-magnitudes sensitivity bounds for total margin at event times t = 0 and t = 1, M-bar in {0.0, 0.5, 1.0, 1.5, 2.0}. Breakdown M-bar at t = 0 is 2.0; at t = 1 is 0.0 (the t = 1 confidence interval includes zero even with no allowed post-period bias). File: `figures/figure5_honestdid_bounds_total_margin.{png,pdf}`.

Figure 6. Propensity-score overlap diagnostic, treated vs. comparison pool. File: `figures/figure6_iptw_propensity_o`

References

1. Centers for Medicare & Medicaid Services. CY 2023 Medicare Hospital Outpatient Prospective Payment System and Ambulatory Surgical Center Payment System Final Rule (CMS 1772-FC): Rural Emergency Hospitals – New Medicare Provider Type. Fact Sheet. 2022. Available from: https://www.cms.gov/newsroom/fact-sheets/cy-2023-medicare-hospital-outpatient-prospective-payment-system-and-ambulatory-surgical-center-1

2. Kim YH, Perry JR, Thompson KW, Pink GH. The First Year of Rural Emergency Hospitals: REHs Serve Relatively Disadvantaged Counties. Findings Brief, NC Rural Health Research Program, Cecil G. Sheps Center for Health Services Research, University of North Carolina at Chapel Hill; 2024.

3. Bipartisan Policy Center. The Rural Emergency Hospital Model: Year Two Progress Report. Washington, DC: Bipartisan Policy Center; 2024.

4. Kaufman BG, Thomas SR, Randolph RK, Perry JR, Thompson KW, Holmes GM, Pink GH. The Rising Rate of Rural Hospital Closures. J Rural Health. 2016;32(1):35–43. doi:10.1111/jrh.12128

5. Lindrooth RC, Perraillon MC, Hardy RY, Tung GJ. Understanding The Relationship Between Medicaid Expansions And Hospital Closures. Health Aff (Millwood). 2018;37(1):111–120. doi:10.1377/hlthaff.2017.0976

6. Holmes GM, Slifkin RT, Randolph RK, Poley S. The effect of rural hospital closures on community economic health. Health Serv Res. 2006;41(2):467–485. doi:10.1111/j.1475-6773.2005.00497.x

7. Durrance CP, Guldi M, Schulkind L. The effect of rural hospital closures on maternal and infant health. Health Serv Res. 2024;59(2):e14248. doi:10.1111/1475-6773.14248

8. Kozhimannil KB, Interrante JD, Carroll C, Sheffield EC, Fritz AH, McGregor AJ, Handley SC. Obstetric Care Access Declined In Rural And Urban Hospitals Across US States, 2010–22. Health Aff (Millwood). 2025;44(7):806–811. doi:10.1377/hlthaff.2024.01552

9. Mullens CL, Johnson PL, Probst JC, Dimick JB, Ibrahim AM, Diaz A. Changes in surgical quality and access after rural hospital closures. Health Aff Sch. 2025;3(5):qxaf089. doi:10.1093/haschl/qxaf089

10. Gujral K, Basu A. Impact of Rural and Urban Hospital Closures on Inpatient Mortality. NBER Working Paper No. 26182. Cambridge, MA: National Bureau of Economic Research; 2019. doi:10.3386/w26182

11. Van Sandt A, Line K, Gruber A, Meier C, Carpenter C, Loveridge S. Rural emergency hospitals: Emerging patterns of adaptation and community perception. J Rural Health. 2026;42(1):e70112. doi:10.1111/jrh.70112

12. Daly SB, You W, Merwin EI. The new Rural Emergency Hospital Designation Program: Will it improve access to care for rural Americans? J Rural Health. 2025;41(4):e70079. doi:10.1111/jrh.70079

13. Schaefer SL, Mullens CL, Ibrahim AM. The Emergence of Rural Emergency Hospitals: Safely Implementing New Models of Care. JAMA. 2023;329(13):1059–1060. doi:10.1001/jama.2023.1956

14. Kerr E, Malone TL, Thompson K, Pink GH, Holmes GM. Did Rural Emergency Hospital Converters Avoid Closure? Ann Emerg Med. 2026 (in press). doi:10.1016/j.annemergmed.2026.02.004

15. Callaway B, Sant'Anna PHC. Difference-in-Differences with multiple time periods. J Econom. 2021;225(2):200–230. doi:10.1016/j.jeconom.2020.12.001

16. Goodman-Bacon A. Difference-in-differences with variation in treatment timing. J Econom. 2021;225(2):254–277. doi:10.1016/j.jeconom.2021.03.014

17. Austin PC, Stuart EA. Moving towards best practice when using inverse probability of treatment weighting (IPTW) using the propensity score to estimate causal treatment effects in observational studies. Stat Med. 2015;34(28):3661–3679. doi:10.1002/sim.6607

18. Chesnaye NC, Stel VS, Tripepi G, Dekker FW, Fu EL, Zoccali C, Jager KJ. An introduction to inverse probability of treatment weighting in observational research. Clin Kidney J. 2022;15(1):14–20. doi:10.1093/ckj/sfab158

19. Sun L, Abraham S. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. J Econom. 2021;225(2):175–199. doi:10.1016/j.jeconom.2020.09.006

20. Chernozhukov V, Chetverikov D, Demirer M, Duflo E, Hansen C, Newey W, Robins J. Double/debiased machine learning for treatment and structural parameters. Econom J. 2018;21(1):C1–C68. doi:10.1111/ectj.12097

21. Holmes GM, Kaufman BG, Pink GH. Predicting Financial Distress and Closure in Rural Hospitals. J Rural Health. 2017;33(3):239–249. doi:10.1111/jrh.12187

22. Malone TL, Pink GH, Holmes GM. An updated model of rural hospital financial distress. J Rural Health. 2025;41(2):e12882. doi:10.1111/jrh.12882

23. Holmes GM, Pink GH, Friedman SA. The financial performance of rural hospitals and implications for elimination of the Critical Access Hospital program. J Rural Health. 2013;29(2):140–149. doi:10.1111/j.1748-0361.2012.00425.x

24. Carroll C, Euhus R, Beaulieu N, Chernew ME. Hospital Survival In Rural Markets: Closures, Mergers, And Profitability. Health Aff (Millwood). 2023;42(4):498–507. doi:10.1377/hlthaff.2022.01191

25. Chatterjee P, Lin Y, Venkataramani AS. Changes in economic outcomes before and after rural hospital closures in the United States: A difference-in-differences study. Health Serv Res. 2022;57(5):1020–1028. doi:10.1111/1475-6773.13988

26. Greenwood-Ericksen M, Kamdar N, Lin P, George N, Myaskovsky L, Crandall C, Mohr NM, Kocher KE. Association of Rural and Critical Access Hospital Status With Patient Outcomes After Emergency Department Visits Among Medicare Beneficiaries. JAMA Netw Open. 2021;4(11):e2134980. doi:10.1001/jamanetworkopen.2021.34980

27. Joynt KE, Chatterjee P, Orav EJ, Jha AK. Hospital Closures Had No Measurable Impact On Local Hospitalization Rates Or Mortality Rates, 2003–11. Health Aff (Millwood). 2015;34(5):765–772. doi:10.1377/hlthaff.2014.1352

28. Rambachan A, Roth J. A More Credible Approach to Parallel Trends. Rev Econ Stud. 2023;90(5):2555–2591. doi:10.1093/restud/rdad018